

Movie-Level Joint Inference for Open-Ended Long-Video QA

Abstract. Open-ended long-video QA requires answers to remain consistent across a film’s characters, relationships, events, and ending state. We treat the movie as the inference unit: a single request combines timestamped subtitles, chronological visual observations, and the complete question set to generate one answer bank. The final SLoMO submission ranked first with 66.36% accuracy and a 3.02 score.

Keywords: long-video QA · movie-level inference · open-ended QA

1 System Overview

MovieQA and TVQA established story-centered video QA with dialogue and visual evidence [3, 8]. LongVideoBench and EgoSchema extend evaluation to sustained multimodal reasoning [5, 9]. MovieChat, TimeChat, Video-ChatGPT, Video-LLaMA, and Qwen-VL advance long-form video language modeling [1, 4, 6, 7, 10]. The SF20K Competition Summary identifies information selection and narrative reasoning as central determinants of story-level QA [2]. Our system treats each film as one context unit and produces one coherent answer bank for its complete question set. Timestamped ASR, chronological visual observations, and all same-movie questions are assembled into one joint request; the answerer returns a question-ID keyed JSON bank for conversion to the submission CSV.

Component	Final configuration
Answer model	GPT-5.6-Luna (<code>gpt-5.6-luna</code>)
API	OpenAI-compatible Chat Completions endpoint
Training	Zero-shot inference, no fine-tuning
Transcript	faster-whisper large-v3-turbo VTT with cue start times
Visual evidence	16 centered uniform chronological frames, 512 px, JPEG quality 82
Generation	Temperature 0, high reasoning effort, 5,000 output tokens, JSON bank
Runtime	Python ≥ 3.10 , OpenAI SDK, HTTPX, OpenCV, and Pillow

2 Inference Protocol

For movie m , shared evidence E_m and the complete question set Q_m map to one bank $\{a_{m,1}, \dots, a_{m,n_m}\}$ in original question order. Audio is transcribed with faster-whisper large-v3-turbo into timestamped VTT cues. We retain cue start

times and the first and last 40,000 characters when needed. For a video with N frames, the i -th observation is $\min(N - 1, \lfloor (i + 0.5)N/16 \rfloor)$ for $i \in \{0, \dots, 15\}$, then resized to a 512-pixel maximum side and JPEG encoded at quality 82. GPT-5.6-Luna receives the transcript, all ordered frames, and all movie questions in one zero-shot request. It silently builds a fact ledger and timeline before returning concise JSON answers with temperature 0, high reasoning effort, and a 5,000-token budget.

3 Evaluation Protocol

Table 1 defines the denominators. Official SLoMO scoring ranks the 538-question leaderboard, while both controlled studies use one fixed Qwen3-VL-32B semantic evaluator. Target16 selects answer-guided locations after answers are withheld from decoding. Dense48 uses label-free uniform coverage.

Study	Scope	Evaluation
Final leaderboard	50 movies, 538 questions	Official 66.36%, Rank 1, score 3.02
Factor study	6 frozen movies, 60 questions	Fixed semantic evaluator, same movie evidence and answerer
Source study	14 movies, 141 questions	Fixed semantic evaluator, shared evidence and questions

Table 1. Evaluation sets and scoring protocol.

4 Results

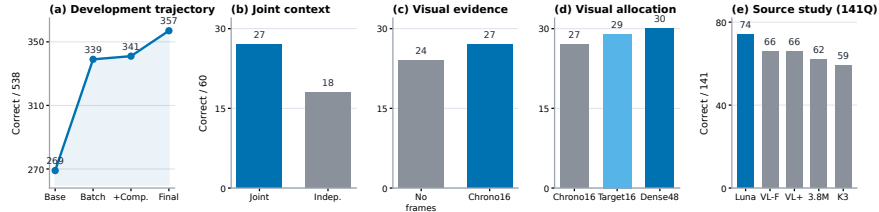


Figure 2. Evidence dashboard. (a) Development score trajectory. (b) Joint context. (c–d) Visual configurations. (e) Source comparison on the common 141-question overlap set. Values are correct answers.

Joint movie answering yields 27/60 versus 18/60 for independent calls. The 141-question source study selects GPT-5.6-Luna at 74 correct answers, above Qwen3-VL Flash, Qwen-VL Plus, Qwen3.8-Max, and Kimi-K3.

References

1. Bai, J., Bai, S., Yang, S., Wang, S., Tan, S., Wang, P., Lin, J., Zhou, C., Zhou, J.: Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond. arXiv preprint arXiv:2308.12966 (2023), <https://arxiv.org/abs/2308.12966>
2. Ghermi, R., Wang, X., Kalogeiton, V., Laptev, I.: SF20K competition 2025: Summary and findings. arXiv preprint arXiv:2605.01496 (2026), <https://arxiv.org/abs/2605.01496>
3. Lei, J., Yu, L., Bansal, M., Berg, T.L.: TVQA: Localized, compositional video question answering. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing (2018), <https://arxiv.org/abs/1809.01696>
4. Maaz, M., Rasheed, H., Khan, S., Khan, F.S.: Video-ChatGPT: Towards detailed video understanding via large vision and language models. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (2024), <https://arxiv.org/abs/2306.05424>
5. Mangalam, K., Akshulakov, R., Malik, J.: EgoSchema: A diagnostic benchmark for very long-form video language understanding. arXiv preprint arXiv:2308.09126 (2023), <https://arxiv.org/abs/2308.09126>
6. Ren, S., Yao, L., Li, S., Sun, X., Hou, L.: TimeChat: A time-sensitive multi-modal large language model for long video understanding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024), <https://arxiv.org/abs/2312.02051>
7. Song, E., Chai, W., Wang, G., Zhang, Y., Zhou, H., Wu, F., Chi, H., Guo, X., Ye, T., Zhang, Y., Lu, Y., Hwang, J.N., Wang, G.: MovieChat: From dense token to sparse memory for long video understanding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024), <https://arxiv.org/abs/2307.16449>
8. Tapaswi, M., Zhu, Y., Stiefelhagen, R., Torralba, A., Urtasun, R., Fidler, S.: MovieQA: Understanding stories in movies through question-answering. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 4631–4640 (2016). <https://doi.org/10.1109/CVPR.2016.501>
9. Wu, H., Li, D., Chen, B., Li, J.: LongVideoBench: A benchmark for long-context interleaved video-language understanding. In: Advances in Neural Information Processing Systems (2024), <https://arxiv.org/abs/2407.15754>
10. Zhang, H., Li, X., Bing, L.: Video-LLaMA: An instruction-tuned audio-visual language model for video understanding. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing: System Demonstrations (2023), <https://arxiv.org/abs/2306.02858>